INCH-POUND

MIL-C-43256D
2 July 1991
SUPERSEDING
MIL-C-43256C
9 June 1989

MILITARY SPECIFICATION

CORD, FIBROUS, POLYESTER, SOLID BRAID

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a solid braided polyester coreless cord.

1.2 Classification. The braided cord shall be of one type, and in the diameters specified in 3.5 (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4020

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SPECIFICATIONS

FEDERAL

UU-T-81 - Tags, Shipping and Stock

MILITARY

MIL-C-3131 - Cordage; Packaging Of

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-595 - Colors Used in Government Procurement

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government Publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 4268 - Testing Fiber Ropes

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Government and contractor purchases. The requirements specified in 3.8 and 3.9 apply only to cord purchased directly by the Government. All other requirements apply both to cord purchased by a contractor as a component for an end item and to cord purchased directly by the Government.

3.3 Standard sample. The cord shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.4 Material. The yarn used in the fabrication of the cord shall be a multifilament, plied or single polyester yarn. When dyed cord is specified, the shade shall be obtained by yarn dyeing or piece dyeing and subsequently the material shall be processed to stabilize the yarn or cord (see 6.5). It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.5 Construction and physical requirements. The cord shall be a solid braid construction without a core. The fabricated cord shall have a compact structure of locked stitches forming a round, even, solid cord and shall be processed to meet the requirements specified in table I when tested as specified in 4.4.3. The character of the cord shall be equal to the standard sample for roundness and firmness (see 6.4).

TABLE I. Physical requirements

Diameter in inches (+1/64)	Number of strands (minimum)	Breaking strength, pounds (minimum)	Length per pound, feet (minimum)	Elongation percent (maximum)
3/32	8	205	360	25
1/8	9	310	220	25
5/32	12	400	140	25
3/16	12	540	99	25
7/32	12	725	71	25
1/4	12	945	53	25

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TABLE I. Physical requirements (cont'd)

Diameter in inches (+1/64)	Number of strands (minimum)	Breaking strength, pounds (minimum)	Length per pound, feet (minimum)	Elongation percent (maximum)
5/16	12	1575	34	25
3/8	12	1925	25	25
1/2	12	2950	14	25

3.6 Color. Unless otherwise specified (see 6.2), the color of the cord shall be natural. When colored cord is specified, the cord shall be dyed Camouflage Green 483, to approximate color chip 34094 of FED-STD-595 (see 6.6).

3.6.1 Colorfastness. The dyed cord shall show fastness to weathering equal to or better than the standard sample or equal to or better than a rating of "good" when tested as specified in 4.4.3.

3.6.2 Matching. The color of the dyed cord shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200K.

3.7 Non-fibrous material (chloroform-soluble material). No extraneous material shall be added for the purpose of weighting the cord. The chloroformsoluble material of the cord shall not exceed 4.0 percent when tested as specified in 4.4.3.

3.8 Put-up. Unless otherwise specified (see 6.2), the cord shall be furnished without knots or splices on spools (reels) in the minimum lengths specified in table II. A plus tolerance of 10.0 percent shall be allowed on the length of spool (reel) specified. Broken lengths will be permitted, but a spool (reel) shall contain not more than three pieces and no piece shall be less than 100 feet in length, or no more than two pieces with no applicable minimum length. The cord shall be wound on the spools (reels) so that each turn and layer is free from entanglement. The ends of the cord shall be heat sealed to prevent fraying.

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TABLE II. Put-up spool (reel) length

Diameter in inches	Length of spool (reel) feet (minimum)
3/32	3000
1/8	3000
5/32	3000
3/16	3000
3/16	100 1/
7/32	2000
1/4	2000
5/16	1000
3/8	1000
1/2	1000

1/ Only one piece permitted.

3.9 Identification. Each spool (reel) of cord shall have a ticket (identification tag) or label attached to it for identification purposes. The ticket shall conform to the requirements for type B, class 1, size 4 or 5 of UU-T-81. The ticket shall be made of not less than 15 point paper stock and shall have a minimum tearing resistance in both directions (total) of 850 grams when tested as specified in UU-T-81. When labels are used, the label shall be attached in such a manner as to remain in place and be clearly legible until all cord has been removed. All ticket or label entries shall be legibly printed, stamped, or typed with water insoluble ink. The ticket or label shall contain the following information:

Stock number
Nomenclature
Size (diameter)
Specification number
Length
Contract number and date
Contractor's name

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3.10 Workmanship. The end item shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2.1 and 4.4.2.2 and tested for the characteristics specified in 4.4.3.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the material conforms to the requirements specified in 3.4.

4.4.2 End item examination.

4.4.2.1 End item visual examination. The cord shall be examined for the defects listed in table III. All defects shall be counted regardless of their proximity to each other. The lot size shall be expressed in units of spools or reels of cord. The sample unit shall be one spool or reel of cord. Ten percent of the total footage, but not less than 100 feet, shall be examined on each spool or reel of cord in the sample. The inspection level shall be II, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for total (major and minor combined) defects.

TABLE III. End item visual defects

Examine	Defect	Classification Major	Minor
Appearance and workmanship	Any cut	101	
	Abraded surface, chafed or damaged	102	
	Kinks or unevenly braided resulting in open places, breaks in continuity of braids, or soft spots	103	
	Loosely braided	104	
	Broken, loose, or projecting strand or end	105	
	Not equal to standard sample for roundness and firmness	106	
Color	Other than specified	107	
	Not uniform		201
Cleanness	Spot or stain clearly visible 1/		202
Identification ticket or label	Omitted, incorrect, illegible, insecurely attached		203
	Ticket or label not as specified		204

1/ At normal inspection distance (approximately 3 feet).

4.4.2.2 Length and winding examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units spools or reels of cord. The sample unit shall be one spool or reel of

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cord. For lots consisting of 500 or fewer units, the sample size shall be 10 and the acceptance number shall be 1. Defects shall be as listed below. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Defects

Length less than specified or more than 10 percent in excess of length specified.

Length of cord on the spool (reel) is less than marked on the ticket or label.

Improperly or not firmly wound resulting in kinking, knotting, entangling, or slippage during unwinding, or otherwise affecting free unhampered unwinding of cord.

Put-up not as specified.

Any end not heat sealed.

Any spool (reel) found to contain more than three pieces.

Any piece less than 100 feet in length (except when there are only two pieces per spool).

Any knot or splice.

4.4.3 End item testing. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table IV shall be followed. When the data in the "Number of determinations" and "Results reported as" columns are not specified in table IV, they shall be as required by the referenced test method. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The lot size shall be expressed in units of spools or reels of cord, as applicable. The sample unit shall be one spool or reel of cord. The sample size shall be in accordance with table V. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more units fail to meet any requirement specified. Tests to determine compliance with specification requirements, including quantity of delivery, may be made under prevailing atmospheric conditions. In cases of dispute, tests shall be made upon material which has reached equilibrium under standard conditions as defined in FED-STD-191.

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TABLE IV. End item tests

Characteristic	Requirement paragraph	Test method	No. of deter. per sample unit	Results reported as
Construction (solid braid without core)	3.5	Visual	1	Pass or fail
Diameter	3.5	ASTM D 4268	-	-
Number of strands	3.5	Visual	1	Pass or fail
Breaking strength and elongation	3.5	6016	-	-
Length per pound	3.5	6004	-	-
Colorfastness to weathering	3.6.1	5671 1/	-	Pass or fail
Nonfibrous material (chloroform-soluble material)	3.7	2611	-	-

1/ The time of exposure shall be 40 standard fading hours.

TABLE V. Sample size

Lot size (spools or reels)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.4.4 Packaging inspection. The sampling and inspection of the preservation, packaging, and container markings shall be in accordance with the requirements of MIL-C-3131, except that the inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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5. PACKAGING

5.1 Levels A, B, and C. Preservation, packing and marking for each level of protection shall be in accordance with the applicable requirements of MIL-C-3131 (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cord is intended for use in tentage, equipage, and other such items, and also for orthopedic and neurosurgical procedures.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Diameter of cord required (see 1.2 and 3.5).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. When color Camouflage Green 483 is required (see 3.6).
f. Put-up, if other than specified (see 3.8).
g. Levels of preservation and packing (see 5.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Twenty-five percent elongation. The maximum elongation requirement of 25 percent has been successfully met by:

a. The use of yarn dyed polyester stabilized prior to braiding.
b. Heat stretching of piece dyed cord.

6.6 Color. Olive green and olive drab colors have been replaced by Camouflage Green 483. Any item which previously required any olive green or olive drab color will now require Camouflage Green 483.

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6.7 Subject term (key word) listing.

Cordage
Equipage
Neurosurgical
Orthopedic
Tent

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Review activities:

Army - MD, MI
DLA - IS

User activities:

Army - ME
Navy - MC

Preparing activity:

Army - GL

(Project 4020-0330)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-43256D	1991 July 2

3. DOCUMENT TITLE
CORD, FIBROUS, POLYESTER, SOLID BRAID

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S. Army Natick RD&E Center	508-651-4532	256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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